Front-Score: A Machine Learning Powered Detection of Cold-Front Signatures.

UNIVERSITY OF KWAZULU-NATAL

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Date: **December 02, 2025**

Goal: To Detect and Quantify Cold Fronts from Station Data.

Aim: To Develop Machine Learning Systems using R.

Objective:
A near-real-time Dashboard.

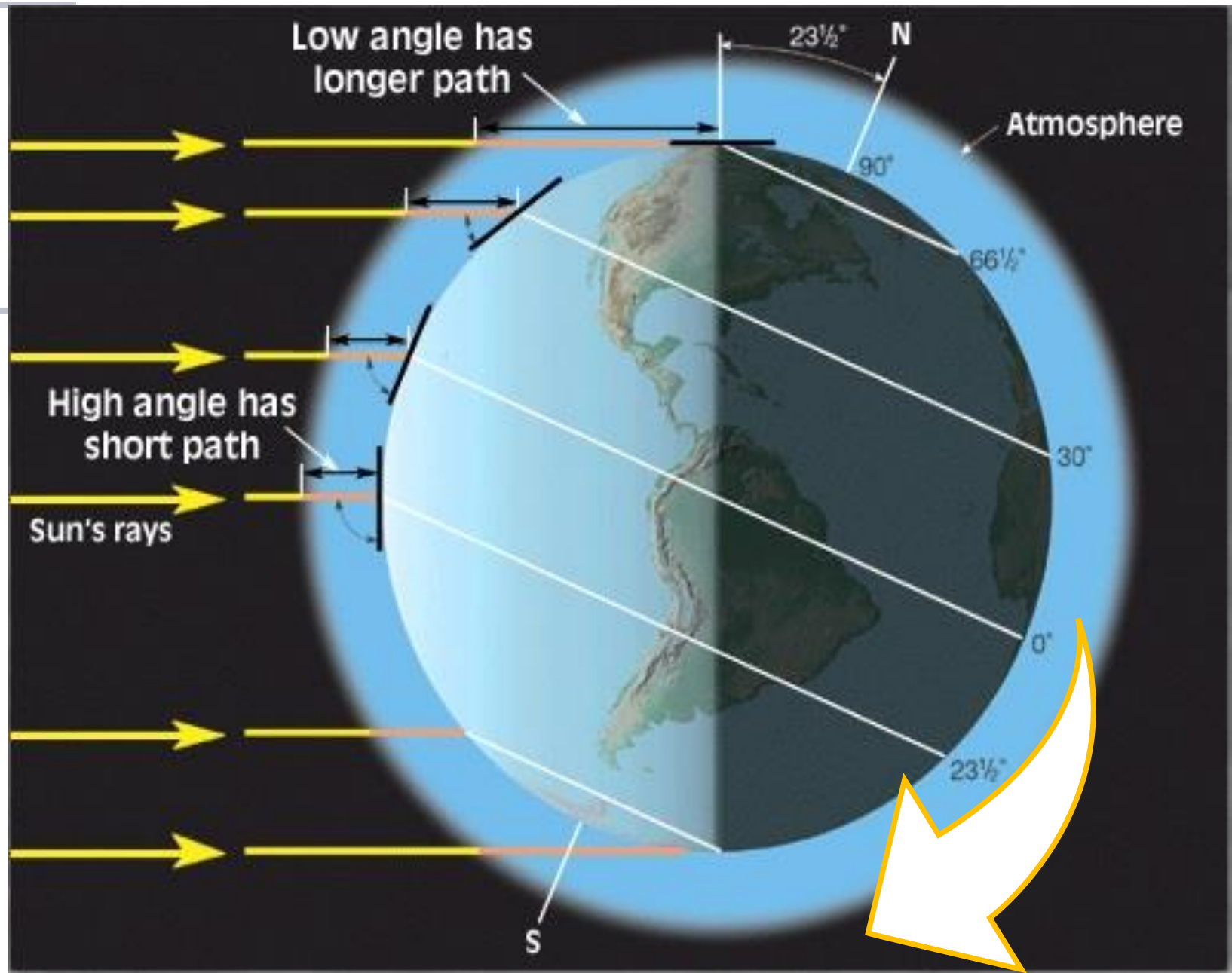
What is a Cold Front?

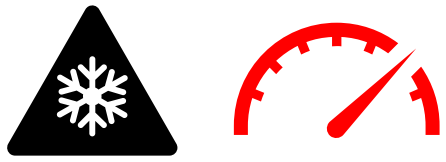
Insolation

Incoming Solar Radiation:

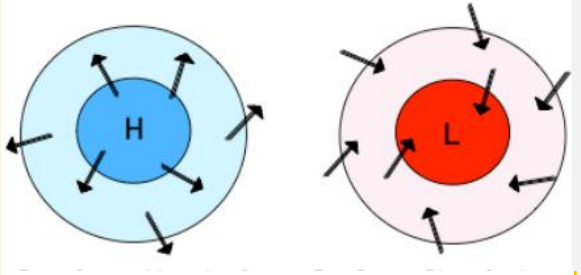
Max at Equator (0° Latitude)

Min at Poles (90° Latitude)

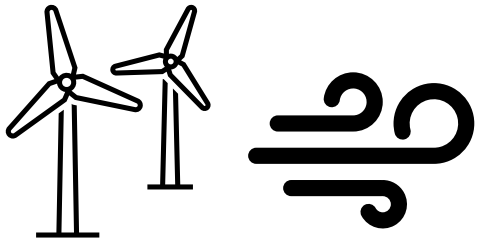




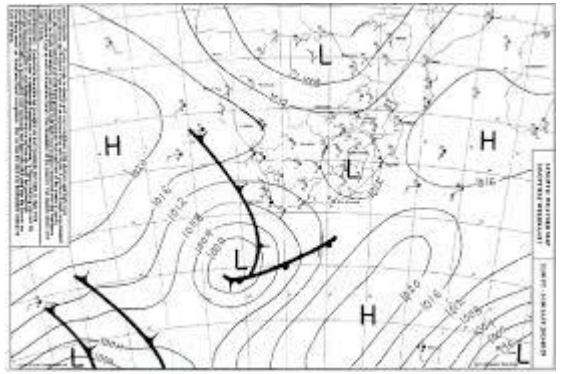
Unequal Heating



Pressure Differences

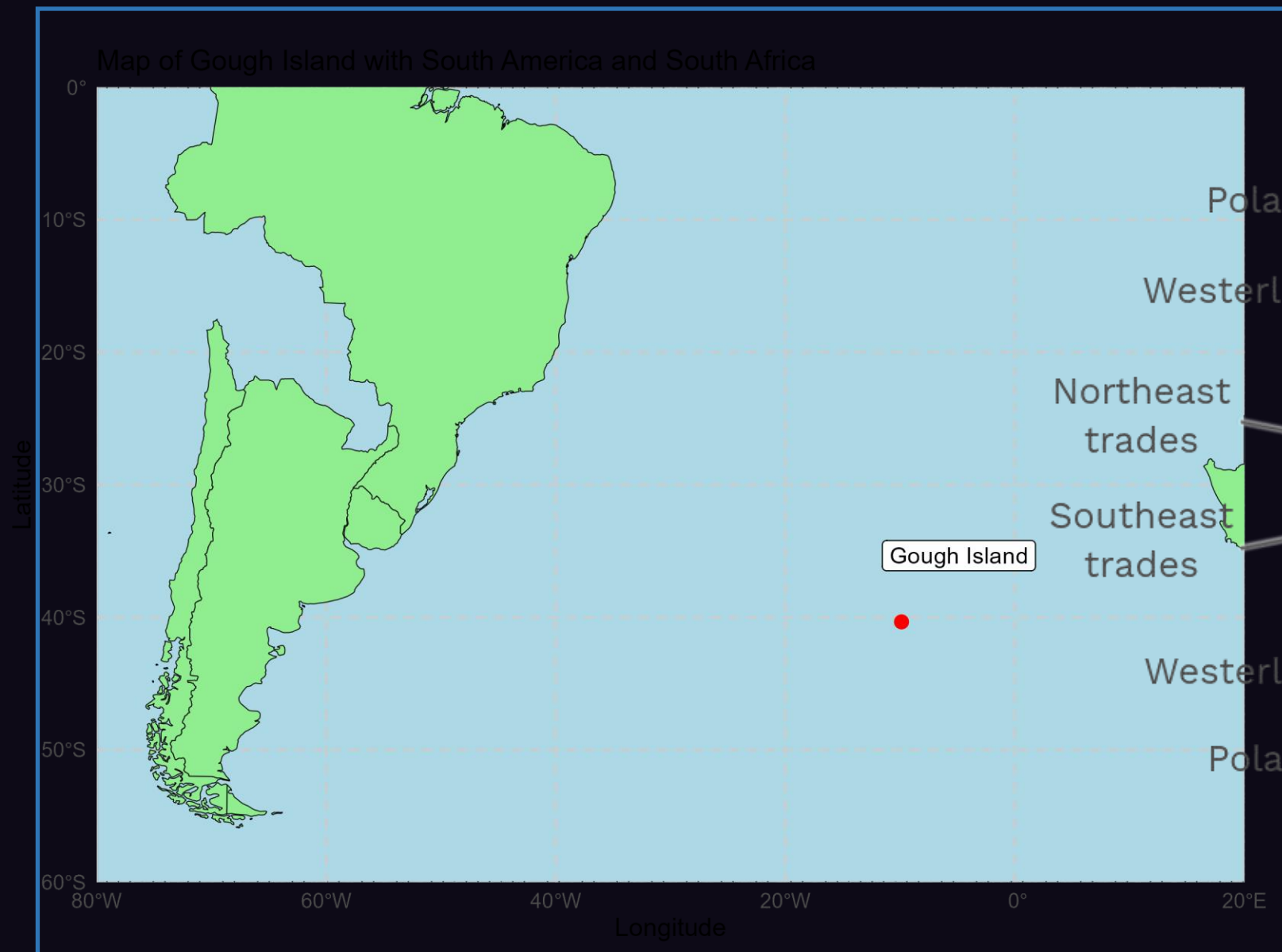


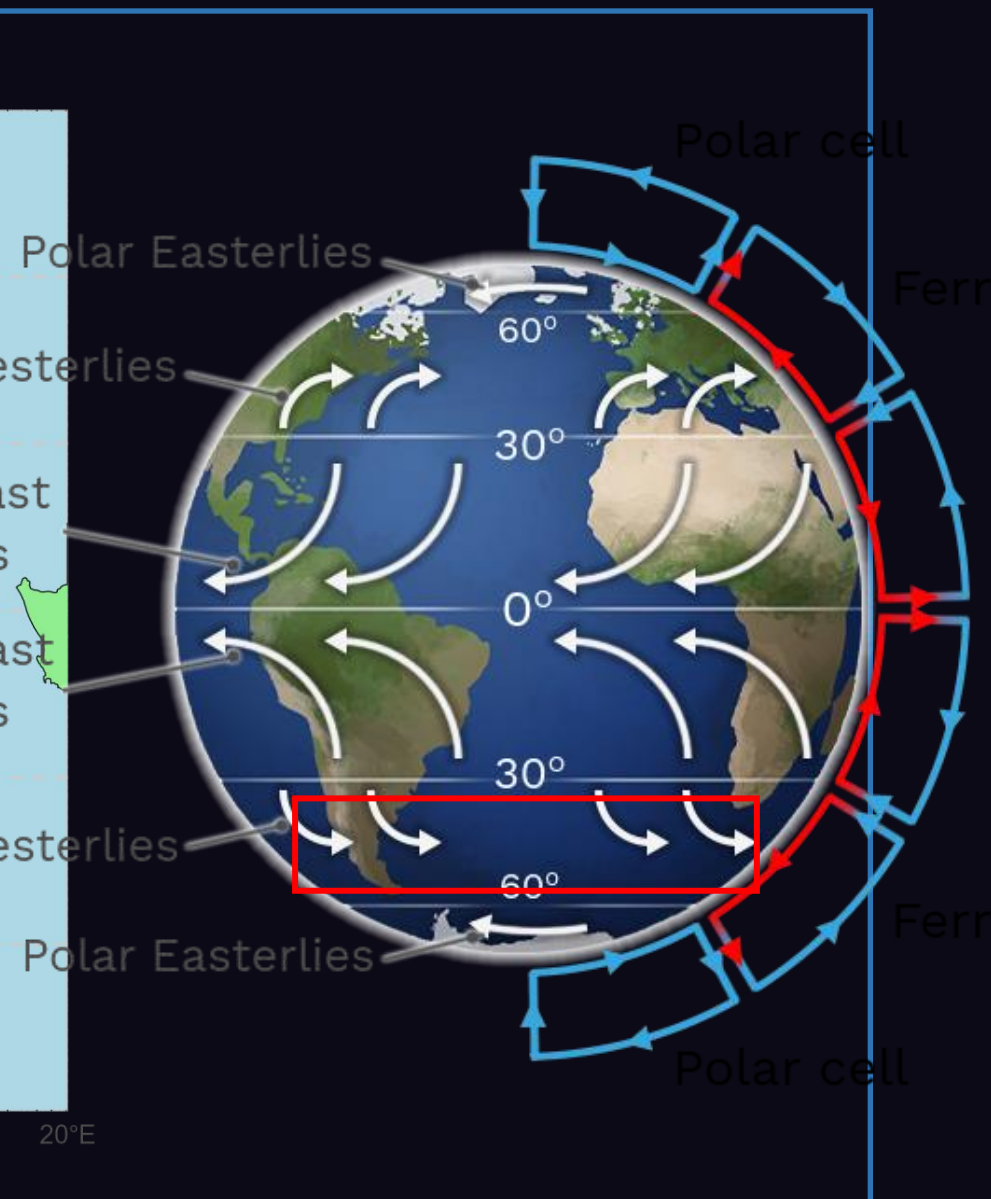
Air Movement



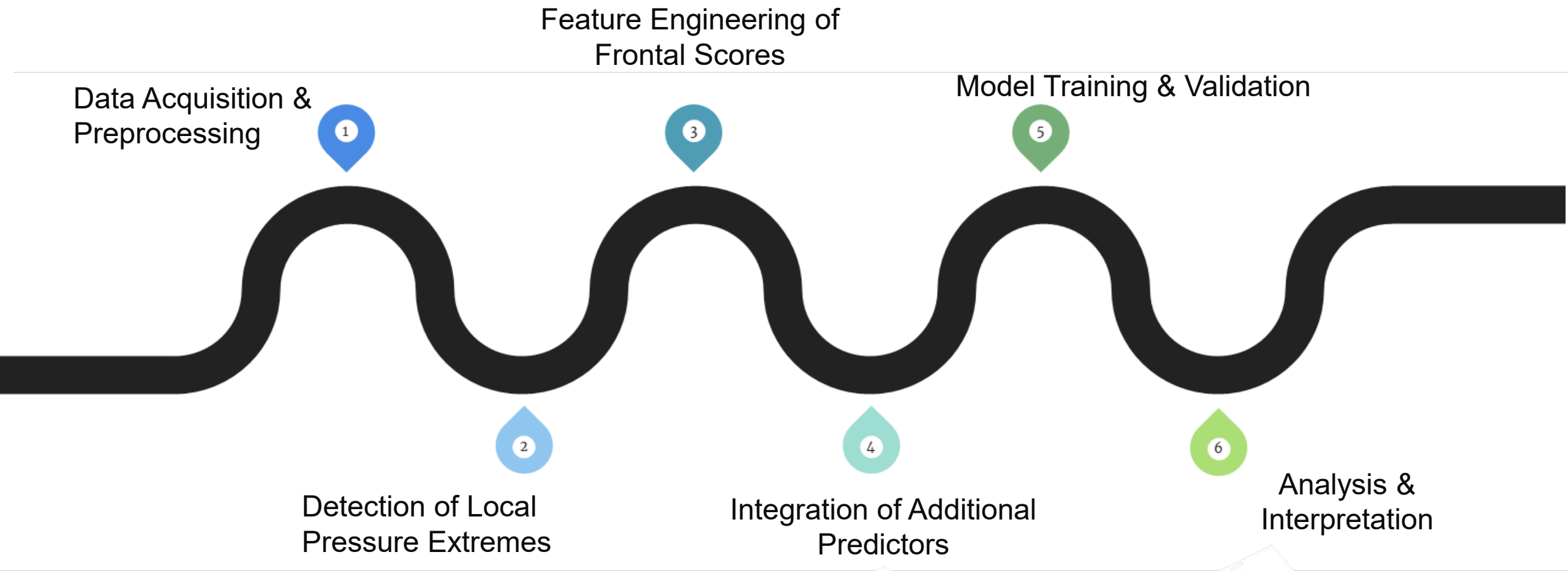
Cold Fronts Form

Why Gough Island?


Map of Gough Island with South America and South Africa
0°
10°S
20°S
30°S
40°S
50°S
60°S
Latitude
80°W
60°W
40°W
20°W
0°
20°E
Longitude
Gough Island


Polar cell
Polar Easterlies
Westerlies
Northeast trades
Southeast trades
Westerlies
Polar Easterlies
60°
30°
0°
30°
60°
Polar cell

Methodology



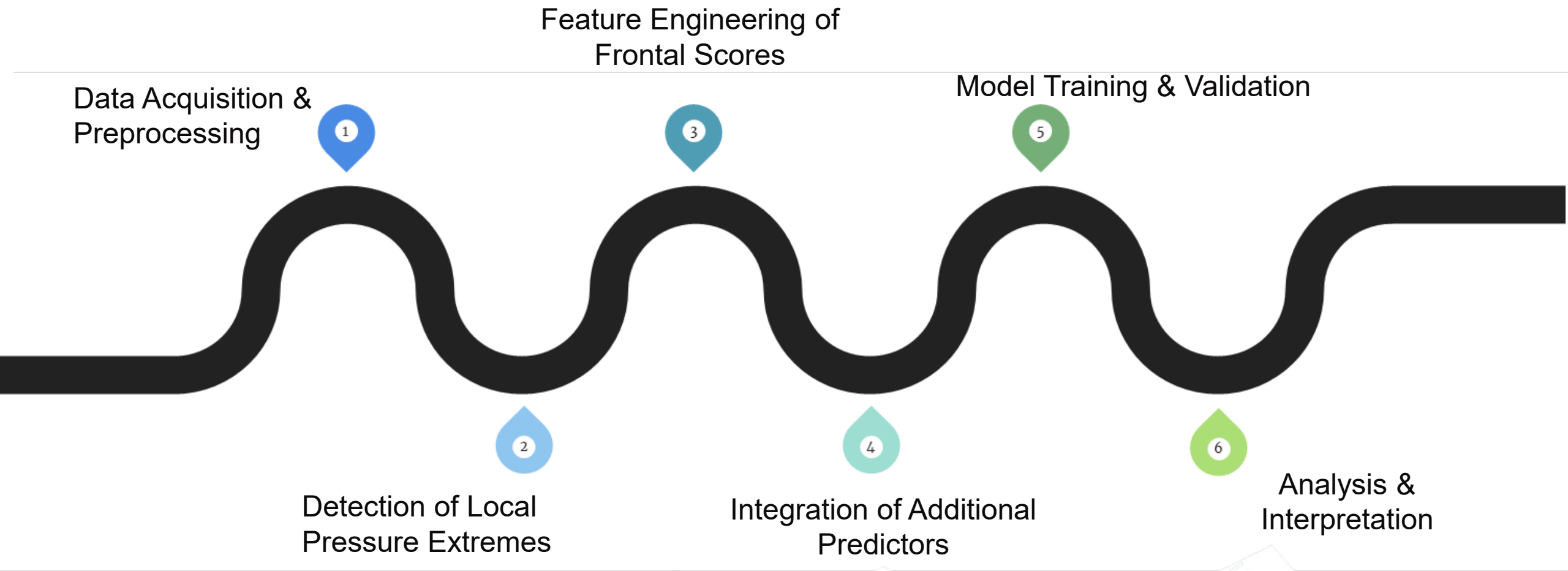
Data Overview

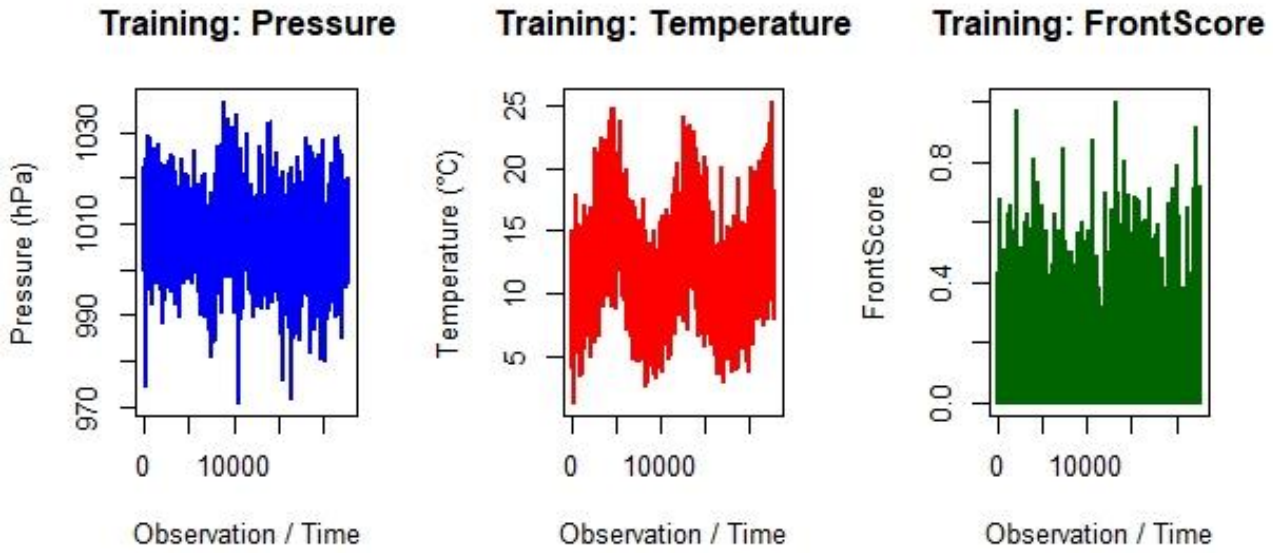
Predictors:

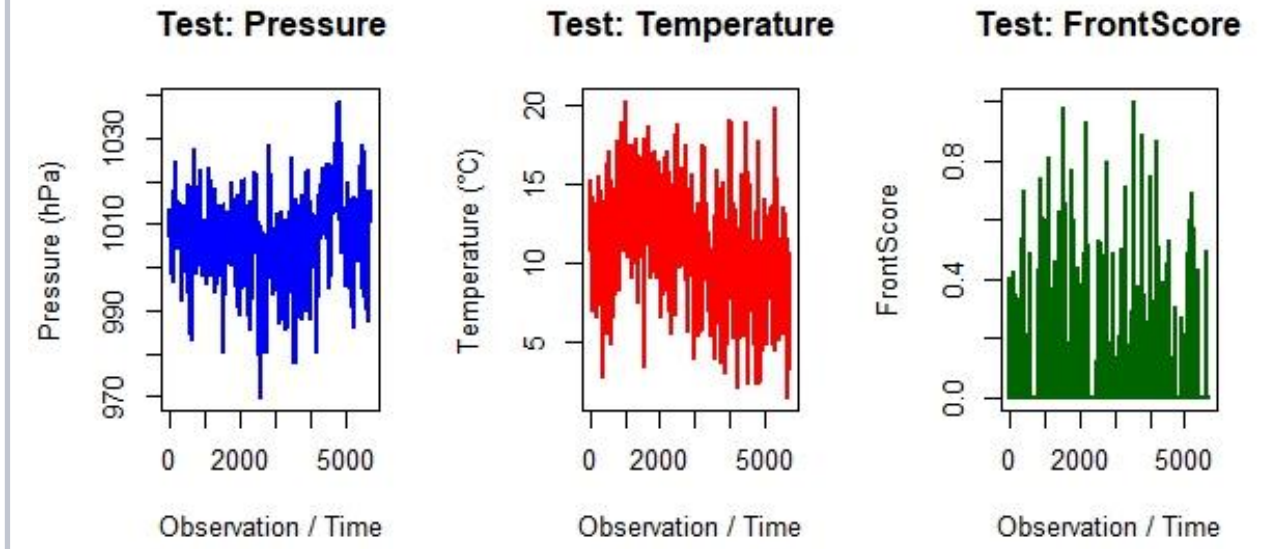
Variable	Type	Units	Description
DateTime	Numeric	(Year, Month, Date, Hour)	Year of observation
Season	Categorical	Season	Meteorological
Day Night	Categorical	(Day/Night)	Time of observation
SST	Numeric	°C	SS TEmp
WindSpeed	Numeric	m/s (or km/h)	Wind speed — scalar value
WindDirection	Circular	Degrees (0–360)	Direction from which the wind is coming (meteorological)
Temperature	Numeric	°C	Air temperature
Rain	Numeric	mm	Precipitation amount
Pressure	Numeric	hPa	Atmospheric pressure
Humidity	Numeric	%	Relative humidity

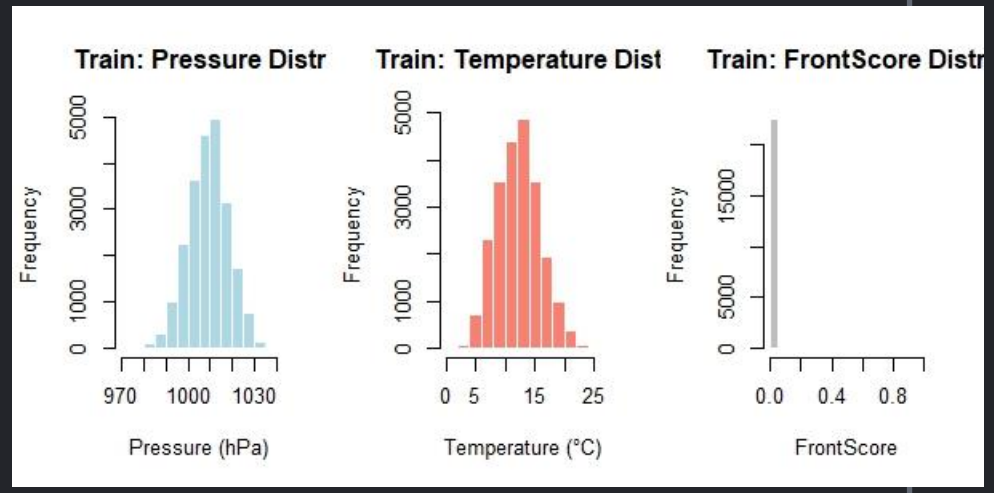
Target:FrontScore, A Composite Score

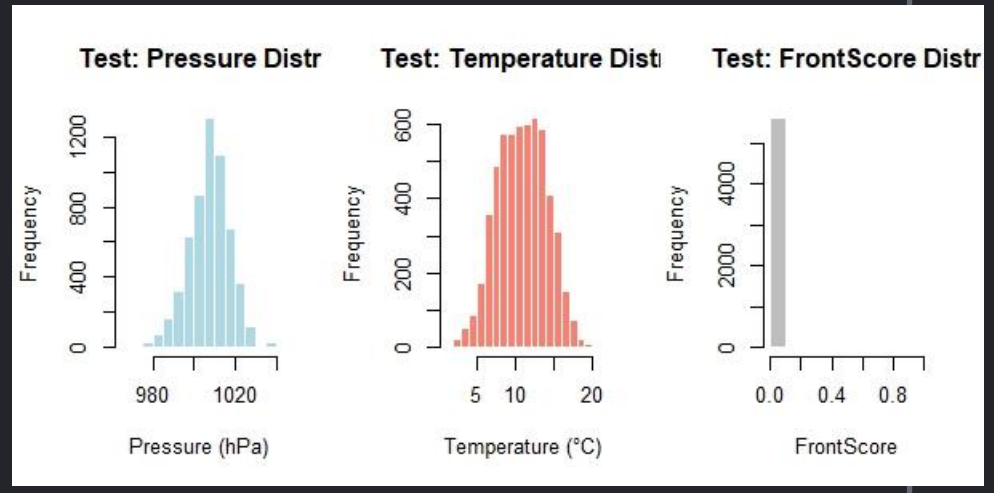
Methodology

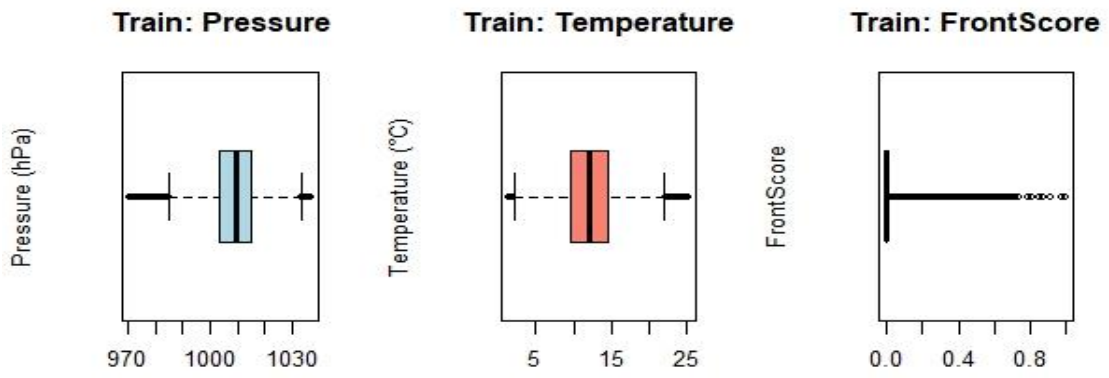


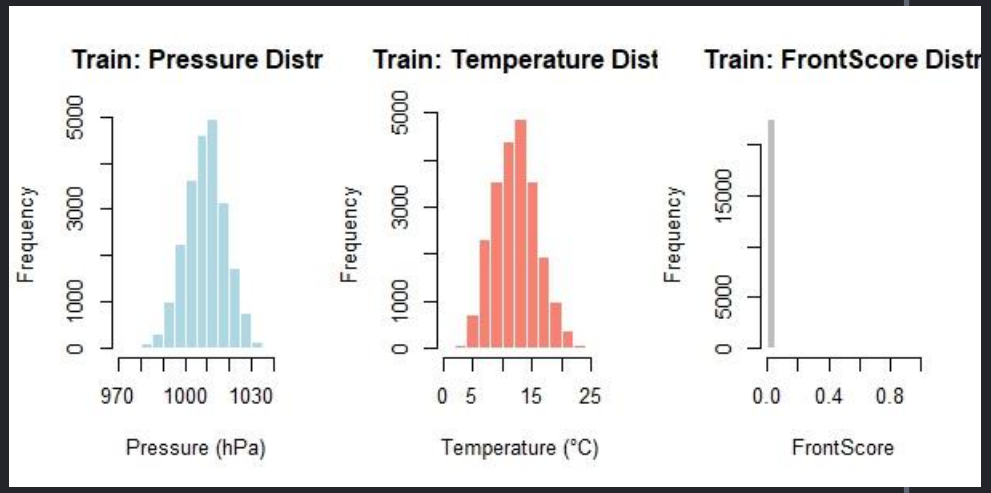

Training: Pressure
Pressure (hPa)
970
990
1010
1030
0
10000
Observation / Time
Training: Temperature
Temperature (°C)
5
10
15
20
25
0
10000
Observation / Time
Training: FrontScore
FrontScore
0.0
0.4
0.8
0
10000
Observation / Time

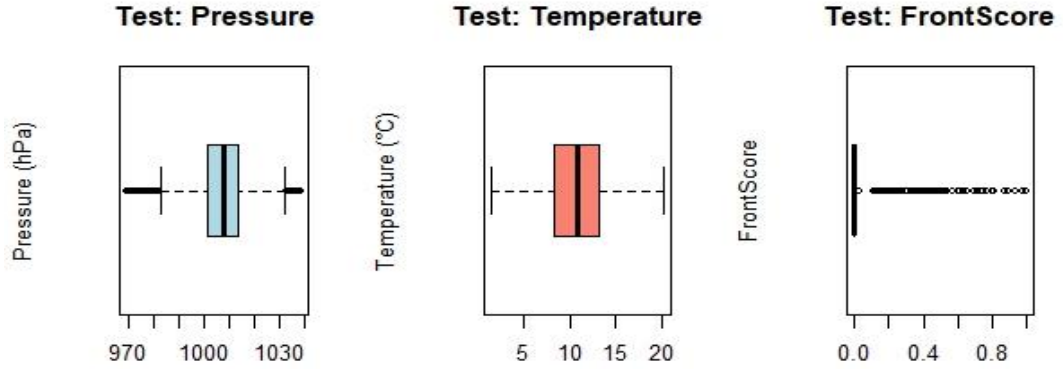

Test: Pressure
Pressure (hPa)
970
990
1010
1030
0
2000
5000
Observation / Time
Test: Temperature
Temperature (°C)
5
10
15
20
0
2000
5000
Observation / Time
Test: FrontScore
FrontScore
0.0
0.4
0.8
0
2000
5000
Observation / Time

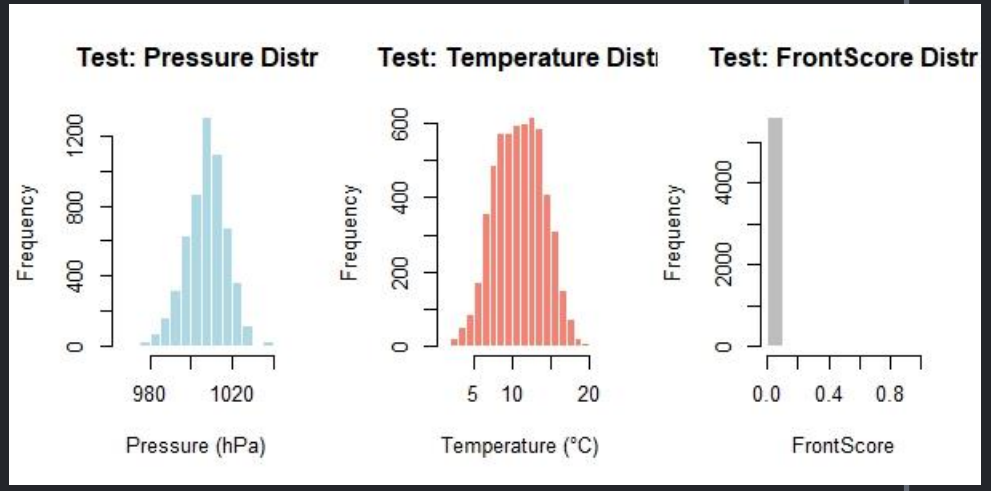

Train: Pressure Distr
Frequency
0
1000
3000
5000
970
1000
1030
Pressure (hPa)
Train: Temperature Dist
Frequency
0
1000
3000
5000
0
5
15
25
Temperature (°C)
Train: FrontScore Distr
Frequency
0
5000
15000
0.0
0.4
0.8
FrontScore


Test: Pressure Distr
Frequency
0
400
800
1200
980
1020
Pressure (hPa)
Test: Temperature Dist
Frequency
0
200
400
600
5
10
20
Temperature (°C)
Test: FrontScore Distr
Frequency
0
2000
4000
0.0
0.4
0.8
FrontScore

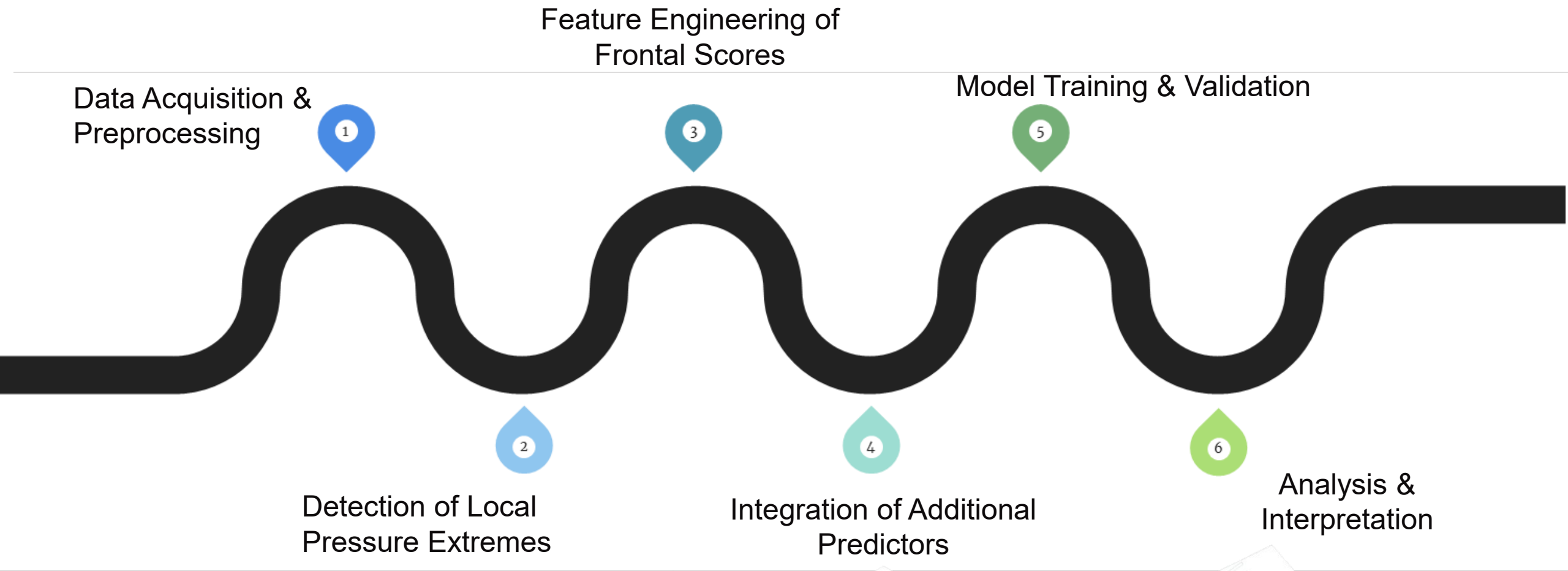

Train: Pressure
Pressure (hPa)
970
1000
1030
Train: Temperature
Temperature (°C)
5
15
25
Train: FrontScore
FrontScore
0.0
0.4
0.8


Train: Pressure Distr
Frequency
0
1000
3000
5000
970
1000
1030
Pressure (hPa)
Train: Temperature Dist
Frequency
0
1000
3000
5000
0
5
15
25
Temperature (°C)
Train: FrontScore Distr
Frequency
0
5000
15000
0.0
0.4
0.8
FrontScore


Test: Pressure
Pressure (hPa)
970
1000
1030
Test: Temperature
Temperature (°C)
5
10
15
20
Test: FrontScore
FrontScore
0.0
0.4
0.8


Test: Pressure Distr
Frequency
0
400
800
1200
980
1020
Pressure (hPa)
Test: Temperature Dist
Frequency
0
200
400
600
5
10
20
Temperature (°C)
Test: FrontScore Distr
Frequency
0
2000
4000
0.0
0.4
0.8
FrontScore

Methodology



Cold Front Passage Behaviour

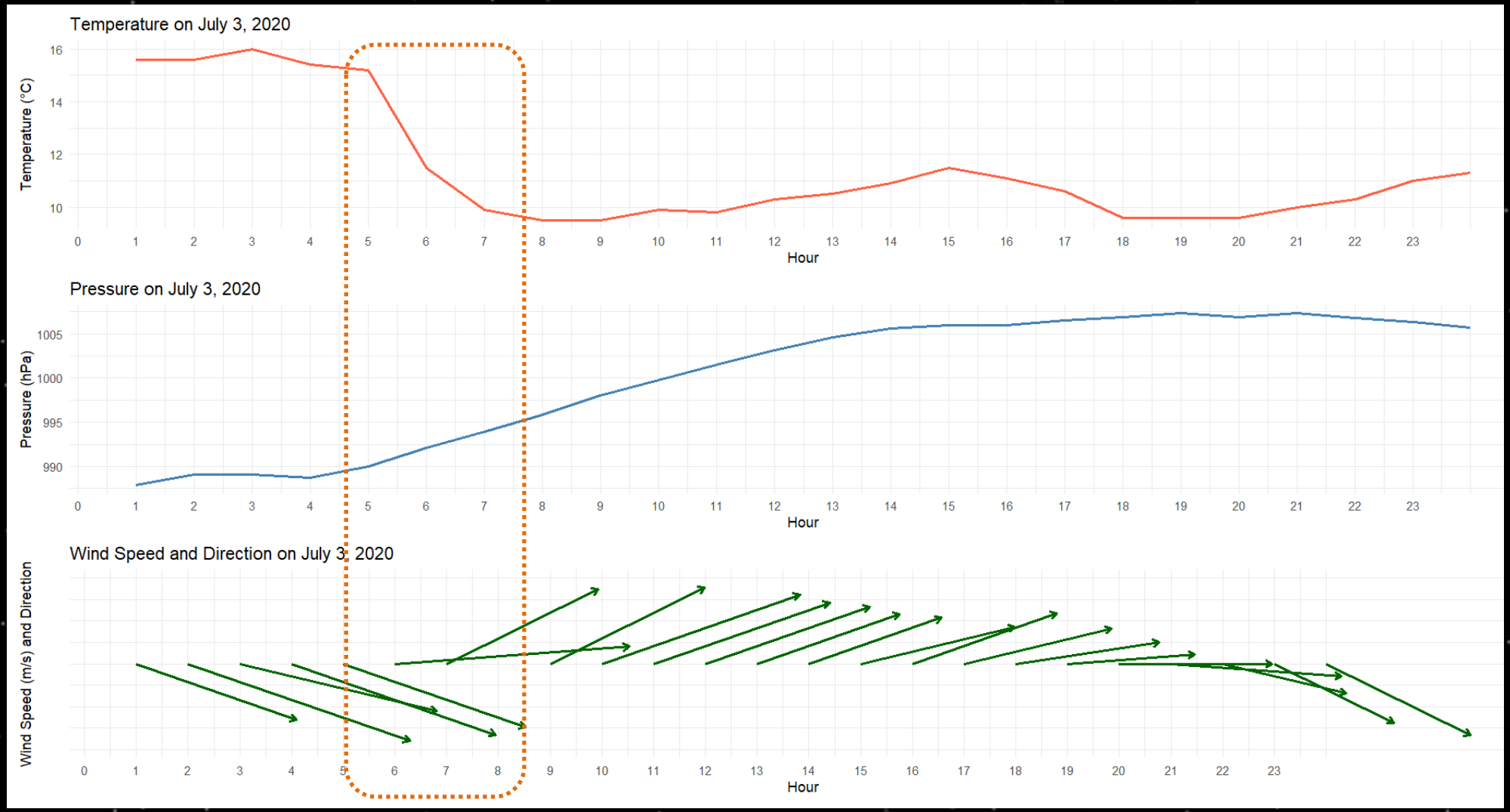
Temperature:
Sudden drop in temperature.

Atmospheric Pressure:

Pressure reaches its lowest point, then starts rising.

Wind:
Shifts abruptly.

Authenticate Cold Front Data



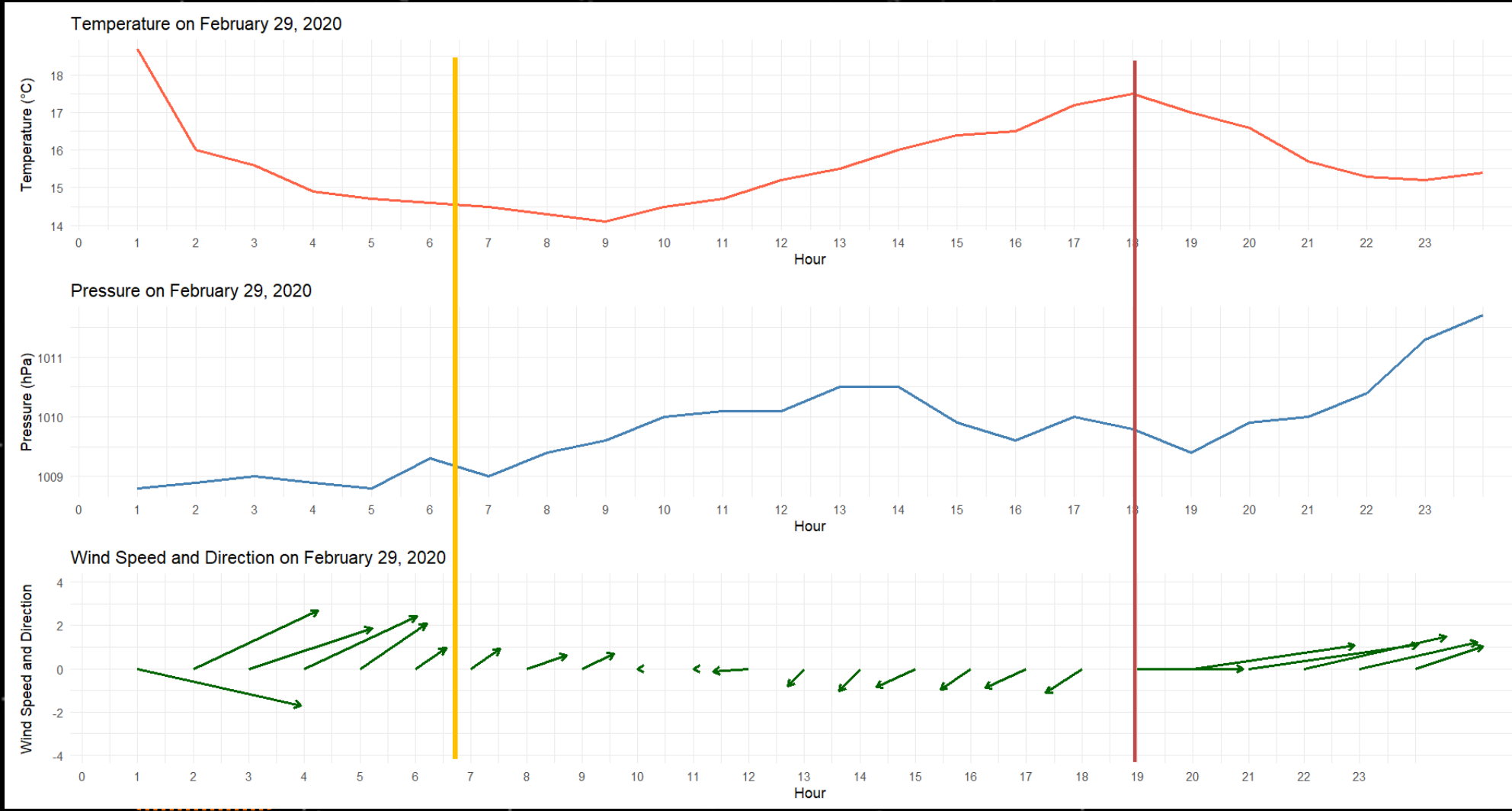
$\Delta T = 6°C$

$\Delta P = 5$ hPA

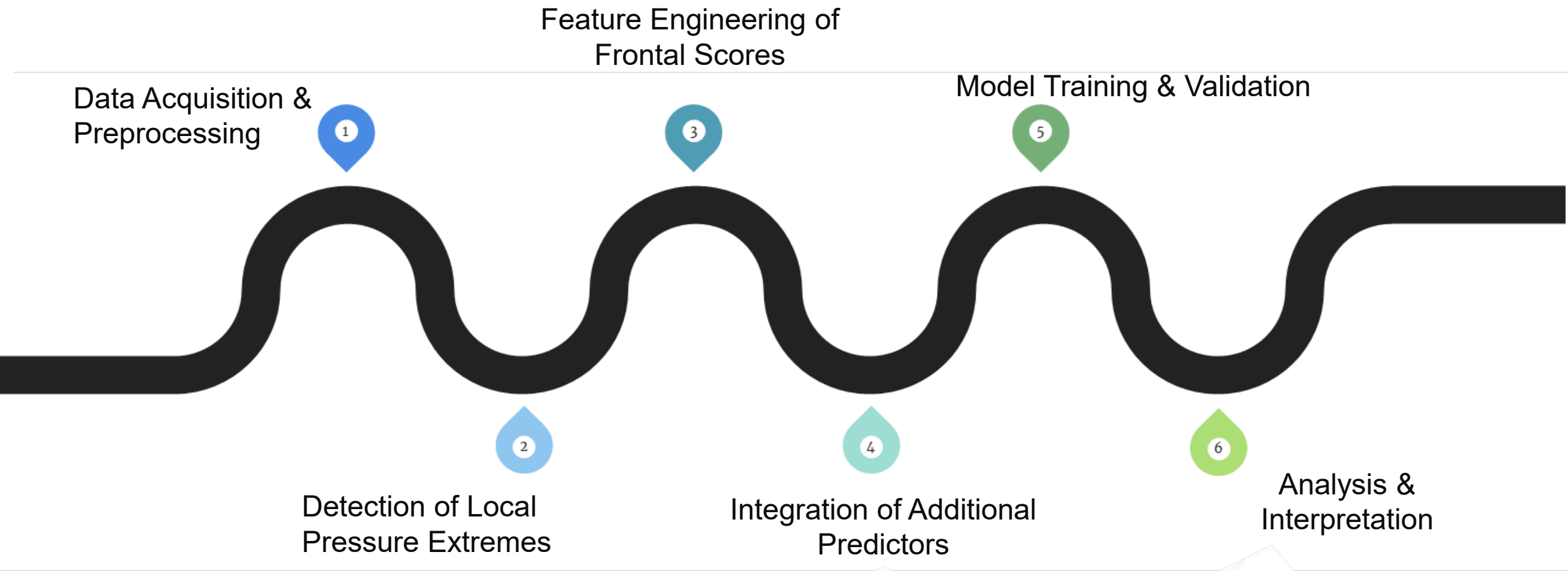
$\Delta\theta = (315 - 225)$

$= 90°$

Diurnal (Day-cycle):



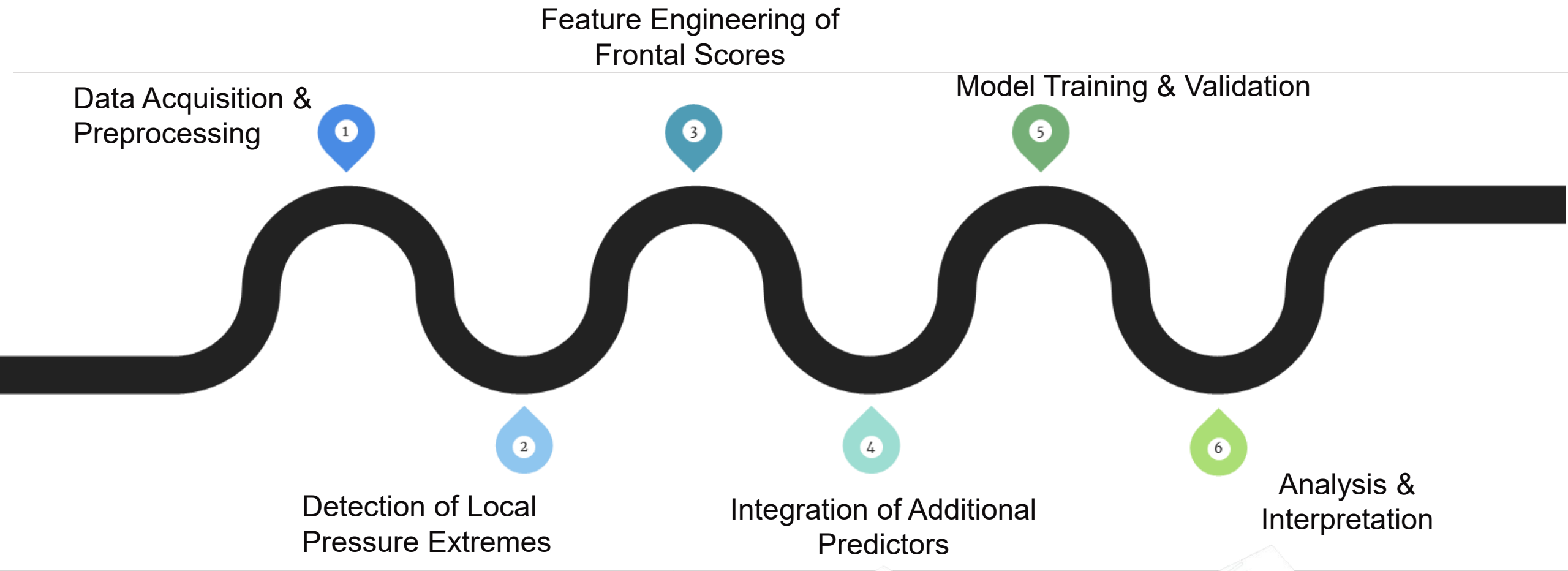
Methodology

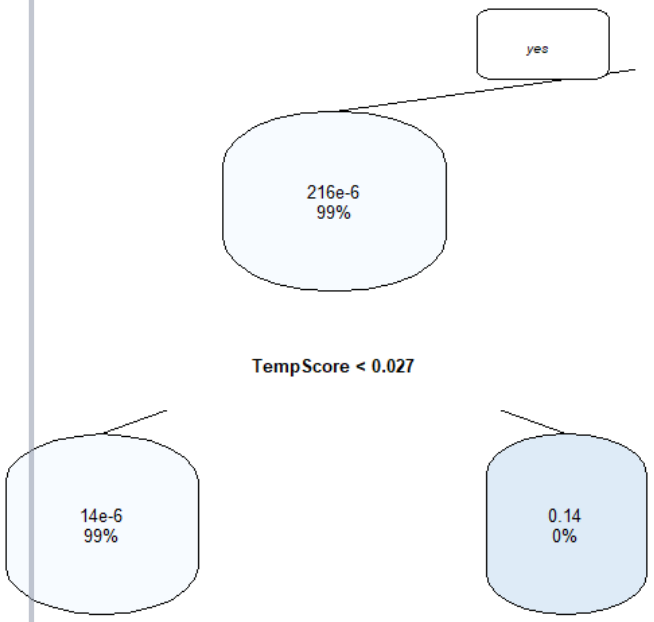


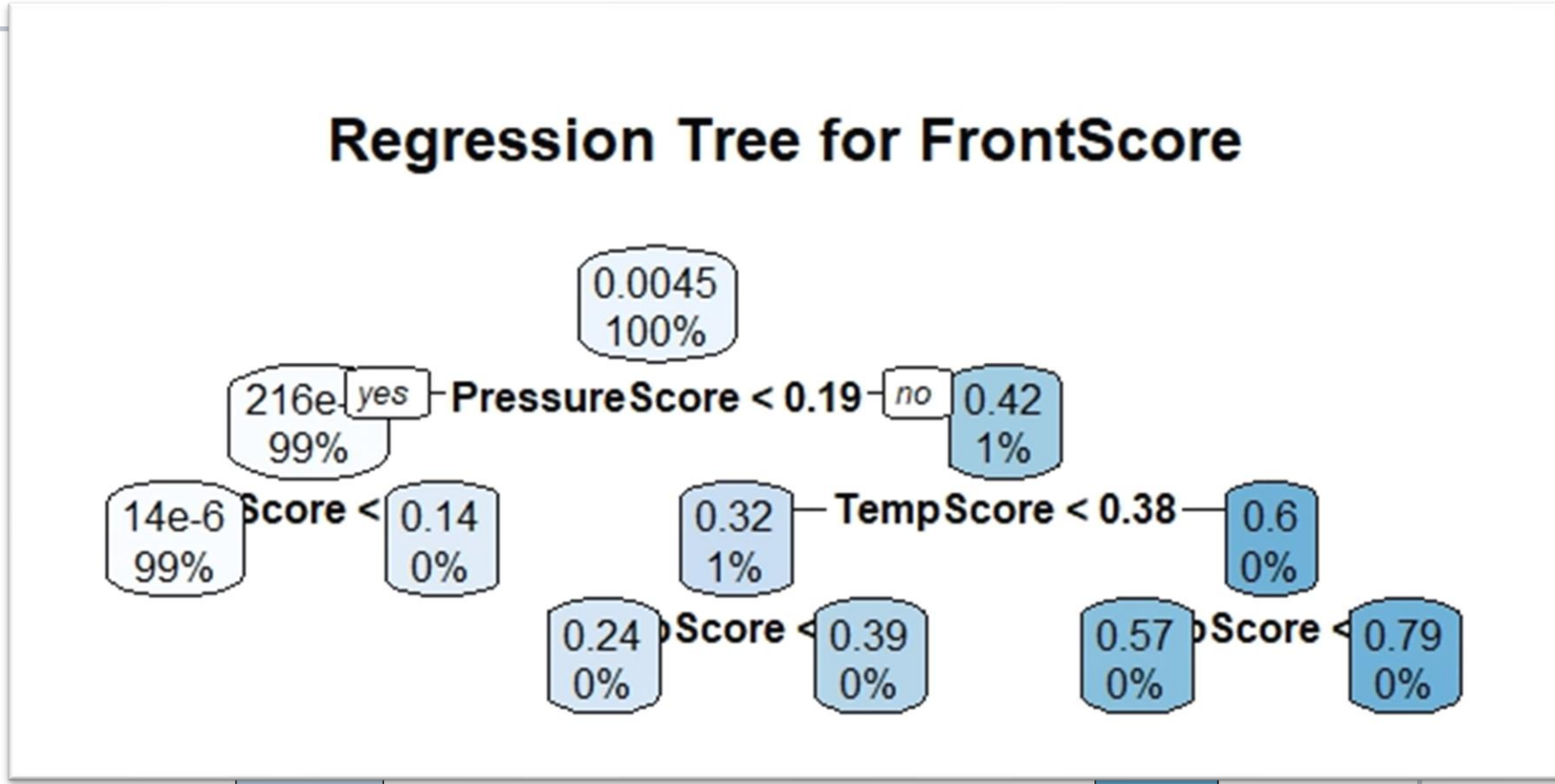
Front Score: 0 to 1

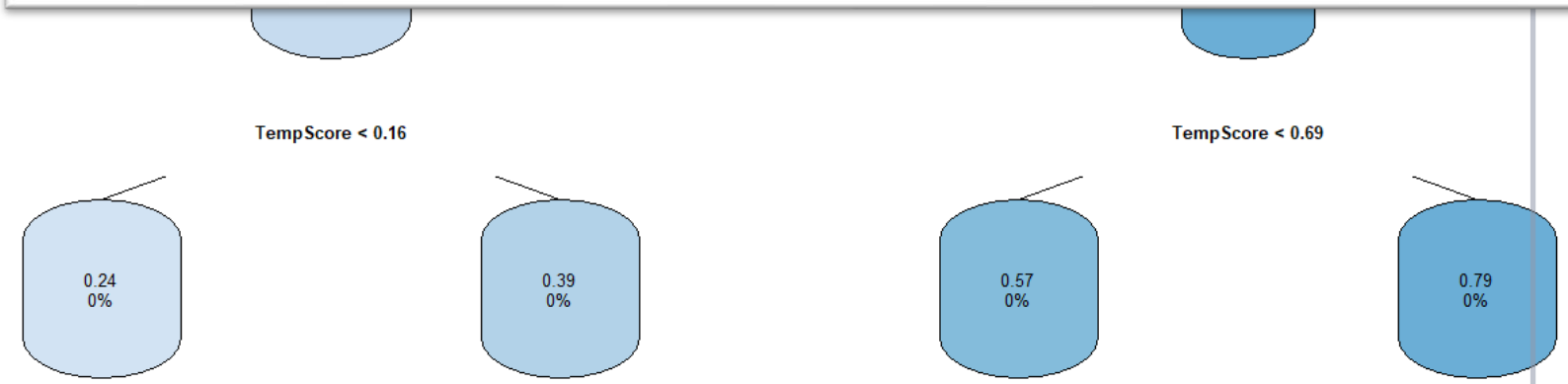
Model	Dataset	MAE	RMSE	Best Hyperparameters / Notes
SVR Linear	Train	0.0265	0.0331	C = 0.1
SVR Linear	Test	0.0271	0.0348	C = 0.1
SVR Radial	Train	0.0309	0.0608	sigma = 0.001, C = 1
SVR Radial	Test	0.0291	0.0907	sigma = 0.001, C = 1
SVR Radial	Train	0.0370	0.0540	sigma = 0.01, C = 5
SVR Radial	Test	0.0713	0.1263	sigma = 0.01, C = 5
SVR Radial	Train	0.0396	0.1135	sigma = 0.01, C = 2
SVR Radial	Test	0.0763	0.2269	sigma = 0.01, C = 2
SVR Poly	Train	0.0242	0.0299	degree = 2, scale = 0.01, C = 5
SVR Poly	Test	0.0235	0.0320	degree = 2, scale = 0.01, C = 5
Ridge Regression	Train	0.0000	0.0001	lambda = 0.001
Ridge Regression	Test	0.0000	0.0004	lambda = 0.001
Lasso Regression	Train	0.0002	0.0011	lambda = 0.001
Lasso Regression	Test	0.0002	0.0013	lambda = 0.001
rpart (Tree)	Train	0.0009	0.0093	-
rpart (Tree)	Test	0.0013	0.0149	-
Random Forest	Train	0.0002	0.0031	-
Random Forest	Test	0.0010	0.0120	-

Methodology




yes
216e-6
99%
TempScore < 0.027
14e-6
99%
0.14
0%


Regression Tree for FrontScore
0.0045
100%
216e
99%
yes
PressureScore < 0.19
no
0.42
1%
14e-6
99%
Score <
0.14
0%
0.32
1%
TempScore < 0.38
0.6
0%
0.24
0%
Score <
0.39
0%
0.57
0%
Score <
0.79
0%


TempScore < 0.16
TempScore < 0.69
0.24
0%
0.39
0%
0.57
0%
0.79
0%

Model	Feature	Notes
Random Forest	TempScore	Most important feature
	PressureScore	Very important
	WindScore	Important
	is_local_min	Important
	pressure	High importance
	temperature	Moderate importance
	windDirection	Moderate importance
	windSpeed	Moderate importance
	humidity	Low importance
	SST_C	Low importance
	Day_Night	Minimal importance
	moon_phase	Negligible
	Season	Negligible
	rain	Negligible
	is_local_max	Not important
Lasso Regression	TempScore	Selected by Lasso
	PressureScore	Selected by Lasso
	WindScore	Selected by Lasso
	Intercept	Baseline

Limitations: Dropped a Lot of Potential Predictors.

Conclusion: Automate Cold Front Analysis.

Way Forward:
A near-real-time dashboard.